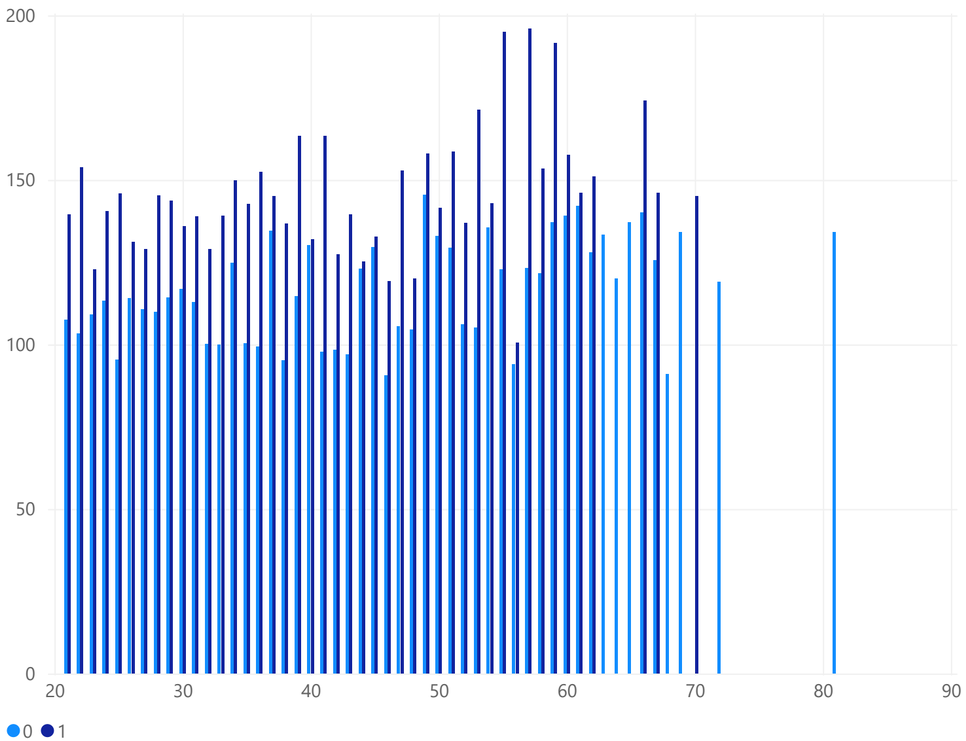
Diabetes Risk Analysis & Patient Overview

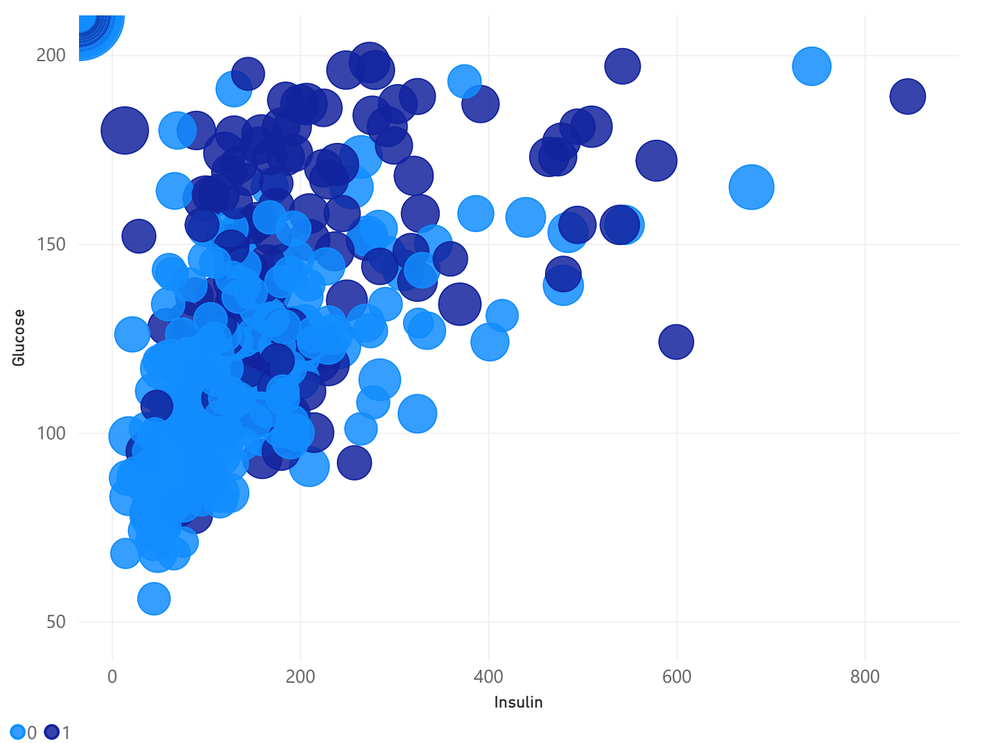
Total Patients
768

Diabetes Rate (%)
0.35

Age
21 81

Outcome
0 1





Data Preparation & Interactive Controls

- Data Cleansing (Power Query): Replaced non-physiological zero values (0) in clinical measurement fields (such as Glucose, BMI, and Insulin) with null values so that missing record placeholders would not skew averages.
- Interactive Slicers: Integrated dynamic filter controls for Outcome and Age, allowing users to slice all visuals across specific age ranges or isolate positive versus negative outcome groups in real time.

Data Visualizations & Clinical Insights

- Average Glucose by Outcome (Clustered Column Chart): Plotted Age on the X-axis against Average Glucose on the Y-axis, categorized by Outcome status. Changed the default aggregation from Sum to Average to accurately reflect clinical baseline shifts across age brackets, highlighting elevated glucose profiles in diabetic patients regardless of age.
- Insulin vs. Glucose Distribution (Scatter Plot): Built a high-density scatter plot with Insulin (X-axis), Glucose (Y-axis), Outcome (Legend), and BMI (Marker Size). Disabled default field summarization (Don't Summarize) to reveal individual patient data points, visually demonstrating clusters of high glucose and insulin resistance across distinct BMI levels.

Key Performance Indicators (KPIs)

- Total Patients: Built a DAX measure (COUNTROWS) displaying 768 total patient records to establish the baseline sample size for the study.
- Diabetes Rate (%): Created a DAX calculation (DIVIDE with CALCULATE on Outcome = 1) showing a 34.9% overall positivity rate, formatting the metric as a clear percentage for instant executive reading.